

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of experience as a Software Developer and Data Engineer, specializing in building scalable systems and robust microservices. Possesses a Master of Science in Computer Science with a strong command of Java/Scala, Python, Spark, SQL, and cloud platforms like AWS and Azure. Proven ability in CI/CD automation, workflow optimization, and developing product features, with expertise in large-scale data platforms and privacy-preserving AI applications.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Built and automated CI/CD pipelines using Jenkins and Azure DevOps for continuous delivery across Azure Cloud and Oracle WebLogic environments, enhancing workflow efficiency.
- Modernized legacy systems by migrating to Log4j 2.x, improving monitoring and error handling to reduce issue resolution time by 20 percent.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles for new product features.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit for comprehensive integration testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, improving overall software development.
- Enhanced system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, ensuring robust software engineering practices.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering team growth.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical product features.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring smooth workflows.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization of product features.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and scalable systems.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, demonstrating infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project) (Python, Hugging Face, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, showcasing micro-services potential.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, enhancing API documentation standards.

SKILLS

- Programming Languages: Proficient in Java/Scala and Python, with extensive SQL expertise for data manipulation and analysis.
- Frameworks and Libraries: Experienced with Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable systems and developing product features.
- Web Services: Skilled in designing and implementing RESTful Services and SOAP APIs, ensuring seamless integration.
- Cloud Platforms: Adept in AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps, ServiceBus) for cloud-native development and infrastructure automation.

- Databases: Strong command of PostgreSQL, Oracle Database, IBM DB2, and RDS, with experience in data warehousing.
- Tools and Technologies: Expertise in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for workflow optimization and CI/CD.
- Concepts: Comprehensive understanding of ETL Pipelines, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Micro-Services, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- Data Analysis and Visualization: Proficient in Tableau and QuickSight for generating actionable insights and presenting complex data.